BINGHAMTON UNIVERSITY
DEPARTMENT OF MATHEMATICAL SCIENCES

Math 147

Elementary Statistics

Course notes, with every exercise worked out

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About these notes

These are the notes I use in the classroom for Math 147, Elementary Statistics, rewritten so that they can be read away from the blackboard. Nothing has been added for the sake of completeness and nothing has been smoothed over: the order of the topics, the examples and the exercises are the ones we actually do in class.

What is new here is that *every exercise is worked out*. In the original handwritten pages many problems were left for the student, and several solutions were sketched in two lines because I was talking while I wrote them. Here each one is carried out step by step, with the reasoning that justifies the step and not only the arithmetic.

Two conventions are worth stating before we start. A box marked REMEMBER holds something you will use again and again, and is worth memorising. A box marked CAREFUL marks a place where students reliably go wrong; I have put one wherever I have seen the same mistake more than twice.

> **HOW TO USE THIS**
>
> Read the statement, close the book, and try the exercise. Only then read the solution. A solution you have read is not a solution you can produce; statistics is learned with a pencil in your hand.

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CHAPTER 1

Getting Ready for Statistics

Statistics is not an arithmetic course, but it does run on arithmetic. Before we meet a single distribution we need to be fluent with four things: proportions and percentages, rounding, reading a formula, and summation notation. This chapter is that warm-up. Every exercise from the first handout appears here, fully worked.

PROPORTION AND PERCENTAGE

A **proportion** is a fraction $\frac{\#1}{\#2}$, where #2 is the whole (the larger number, the total) and #1 is the part of it you care about. It always lands between 0 and 1. A **percentage** is the same number with the decimal point moved two places to the right:

$$\text{percentage} \;=\; \text{proportion} \times 100\%.$$

The whole difficulty in these problems is deciding what plays the role of #2. Decide that first, and the arithmetic follows.

1.1 Proportions and percentages

Exercise 1

Sage pays \$550 per month for rent and her monthly salary is \$1876.

(a) What proportion of her monthly salary is her rent? (This answer will be between 0.000 and 1.000.)

(b) What percentage of her monthly salary is her rent? (This answer will be between 0.0% and 100.0%.)

Solution. The whole here is the salary, because the rent is a part of it. So #2 $= 1876$ and #1 $= 550$.

(a) The proportion is

$$\frac{550}{1876} = 0.29317\ldots \approx 0.293.$$

(b) Multiplying by 100,

$$0.29317\ldots \times 100 = 29.317\ldots\,\% \approx 29.32\%.$$

Rent eats a little under a third of what she earns.

(a) 0.293 (b) 29.32%

Exercise 2

At Binghamton High School there are 295 seniors who entered Binghamton schools in the fifth grade or lower, and 798 seniors who entered in the sixth grade or higher. What percentage of the seniors entered Binghamton schools in the fifth grade or lower?

Solution. The whole is *all* the seniors, and that number is not given directly: we have to build it.

$$\#2 = 295 + 798 = 1093, \qquad \#1 = 295.$$

Then

$$\frac{295}{1093} = 0.26989\ldots \approx 26.99\%.$$

26.99%

Exercise 3

In a particular arts and humanities college, the following table gives the numbers of men and women in various majors.

	Men	**Women**	**Total**
Anthropology	83	69	152
History	102	110	212
Music	53	37	90
Total	238	216	454

(a) What percentage of the women are anthropology majors?

(b) What percentage of the music majors are men?

Solution. Read each question backwards: the group named *after* the word "of" is the whole.

(a) "of the women" ⇒ the whole is all 216 women, and the part is the 69 women in anthropology:

$$\frac{69}{216} = 0.31944\ldots \approx 31.94\%.$$

(b) "of the music majors" ⇒ the whole is the 90 music majors, and the part is the 53 men among them:

$$\frac{53}{90} = 0.58888\ldots \approx 58.89\%.$$

(a) 31.94% (b) 58.89%

CAREFUL

Notice how different (a) and (b) are, even though both use numbers from the same table. $\frac{69}{216}$ and $\frac{69}{152}$ answer two different questions: "what percentage of the women study anthropology" and "what percentage of the anthropology students are women".

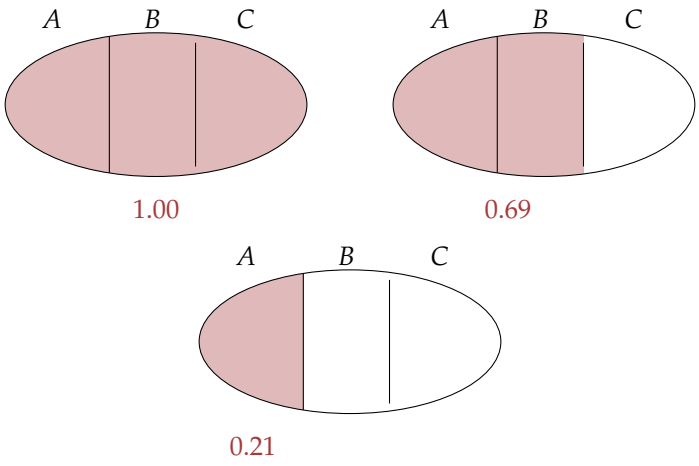
> Choosing the wrong denominator is the single most common mistake in this chapter.

1.2 Areas as proportions

Proportions do not have to come from counting people. An area can be a proportion too, and this is exactly how we will read probability later in the course.

Exercise 4

A figure is divided into three non-equal pieces by lines. The three pictures below are all of the same figure with the same division lines. Different areas are shaded and labelled on each. Find the areas of the portions labelled B and C.



Solution. Name the three unknown areas A, B, C. Each picture gives one equation:

$$A + B + C = 1.00, \qquad A + B = 0.69, \qquad A = 0.21.$$

Now subtract, from the smallest piece of information upwards. Taking the first two,

$$C = (A + B + C) - (A + B) = 1.00 - 0.69 = 0.31,$$

and taking the last two,

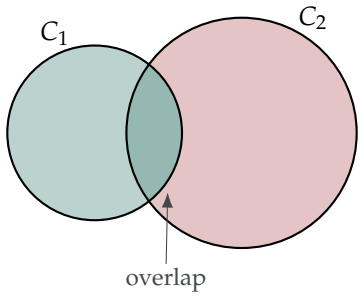
$$B = (A + B) - A = 0.69 - 0.21 = 0.48.$$

As a check, $0.21 + 0.48 + 0.31 = 1.00$, as it must be.

$B = 0.48, \qquad C = 0.31$

Exercise 5

Two circles overlap as sketched below. The area of circle 1 is 0.48 and the area of circle 2 is 0.57.



(a) Can you figure out the shaded area or do you need more information? What would that be?

(b) The overlapping area of the two circles is 0.16. Can you now find the shaded area?

Solution. **(a)** No, not yet. In Exercise 4 the pieces did not overlap, so areas simply added up. Here the two circles share a region, and if we write $0.48 + 0.57$ that shared region has been counted *twice*. What is missing is exactly the area of the overlap.

(b) With the overlap equal to 0.16, count circle 1 without its shared part, and then all of circle 2:

$$\text{shaded} = \underbrace{(0.48 - 0.16)}_{\text{only } C_1} + \underbrace{0.57}_{C_2} = 0.32 + 0.57 = 0.89.$$

Equivalently, add everything and remove the double count: $0.48 + 0.57 - 0.16 = 0.89$.

0.89

OVERLAPPING PIECES

When two regions overlap,

$$\text{area}(C_1 \text{ or } C_2) = \text{area}(C_1) + \text{area}(C_2) - \text{area}(C_1 \text{ and } C_2).$$

This is the addition rule, and we will meet it again word for word when we do probability.

1.3 Rounding and the calculator

HOW TO ROUND

To round to a given decimal place, look at the *next* digit to the right. If it is 5 or more, add one to the digit you are keeping; if it is 4 or less, leave it alone. Then drop everything after it.

Exercise 6

Round 57.39283 to the nearest thousandth.

Solution. "Thousandth" means three decimal places, so we keep 57.392 and look at the fourth decimal:

$$57.392\,\underline{8}3.$$

That digit is $8 \geq 5$, so the third decimal goes up from 2 to 3.

57.393

Exercise 7

Round 0.02397 to the nearest hundredth.

Solution. "Hundredth" means two decimal places, so we keep 0.02 and look at the third decimal:

$$0.02\,\underline{3}97.$$

That digit is $3 \leq 4$, so the second decimal stays as it is. The 97 that follows is irrelevant — only the first discarded digit decides.

0.02

CAREFUL

In Exercise 7 it is tempting to look at the 9 and round up, reasoning that 0.02397 is "almost 0.024". Rounding is decided by one digit only, the first one you drop. Chaining the rounding ($0.02397 \to 0.0240 \to 0.024 \to 0.02$) can give a different answer and is wrong.

Exercise 8

Use your calculator to find $\sqrt{21}$, correct to 3 decimal places.

Solution. The calculator gives $\sqrt{21} = 4.58257569\ldots$. Keeping three decimals we look at the fourth, which is 5, so the third goes up from 2 to 3.

$\sqrt{21} \approx 4.583$

Exercise 9

Use your calculator to find 18^2.

Solution. Squaring means multiplying the number by itself: $18^2 = 18 \times 18 = 324$.

324

Exercise 10

When a scientific calculator shows the answer for a computation as 5.38E − 04, what number does that represent?

Solution. The display 5.38E − 04 is the calculator's way of writing scientific notation:

$$5.38\text{E} - 04 \;=\; 5.38 \times 10^{-4}.$$

A negative exponent moves the decimal point to the *left*, as many places as the exponent says — here four places:

$$5.38 \to 0.538 \to 0.0538 \to 0.00538 \to 0.000538.$$

$5.38\text{E}-04 = 0.000538$

1.4 Inequalities and interval notation

Exercise 11

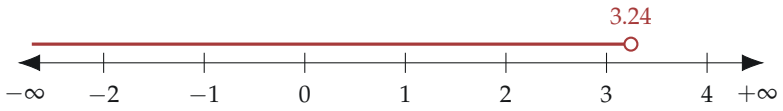
Consider the mathematical statement $x < 3.24$.

(a) How would you read that statement in words?

(b) Give the interval notation for all the numbers that make that statement true. Draw it on a number line.

Solution. **(a)** "x is less than 3.24."

(b) Every number strictly to the left of 3.24 works, and 3.24 itself does not. In interval notation that is $(-\infty, 3.24)$. The round bracket, and the hollow dot in the picture, both say "this endpoint is not included".



$(-\infty, 3.24)$

Exercise 12

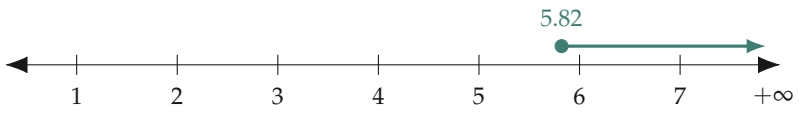
Consider the mathematical statement $y \geq 5.82$.

(a) How would you read that statement in words?

(b) Give the interval notation for all the numbers that make that statement true. Draw it on a number line.

Solution. **(a)** "y is greater than or equal to 5.82."

(b) Now 5.82 *is* allowed, because of the "or equal to". The square bracket, and the solid dot, both say "this endpoint is included".



$[5.82, +\infty)$

BRACKET OR PARENTHESIS

Use a square bracket [ ] when the endpoint belongs to the set ($\leq$ or $\geq$), and a round parenthesis ( ) when it does not ($<$ or $>$). Infinity never belongs to anything, so it always takes a parenthesis.

1.5 Solving for a letter

UNDOING OPERATIONS

To isolate a letter, undo what is being done to it, in reverse order. Each operation is undone by its opposite:

$$+ \longleftrightarrow -, \qquad \times \longleftrightarrow \div, \qquad (\,)^2 \longleftrightarrow \sqrt{\ }.$$

Whatever you do to one side of the equals sign, you do to the other.

Exercise 13

Solve for x: $5 + 3x = 18$.

Solution. The 5 is added, so subtract it from both sides; the 3 multiplies, so divide by it.

$$5 + 3x = 18 \implies 3x = 18 - 5 = 13 \implies x = \frac{13}{3}.$$

As a decimal, $x = 4.333\ldots \approx 4.333$. Check: $5 + 3 \cdot \frac{13}{3} = 5 + 13 = 18$. ✓

$x = \frac{13}{3} \approx 4.333$

Exercise 14

Solve for y: $7y - 8 = 24 + 26y$.

Solution. The letter appears on both sides, so first gather the y's on one side and the numbers on the other.

$$7y - 8 = 24 + 26y \implies 7y - 26y = 24 + 8 \implies -19y = 32 \implies y = -\frac{32}{19}.$$

As a decimal, $y = -1.6842\ldots \approx -1.684$. Check: $7(-1.6842) - 8 = -19.789$ and $24 + 26(-1.6842) = -19.789$. ✓

$y = -\frac{32}{19} \approx -1.684$

Exercise 15

Solve for r: $\dfrac{7}{5} = \dfrac{6}{r}$.

Solution. Two fractions set equal to each other: cross-multiply.

$$\frac{7}{5} = \frac{6}{r} \implies 7r = 5 \cdot 6 = 30 \implies r = \frac{30}{7}.$$

As a decimal, $r = 4.2857\ldots \approx 4.286$.

$r = \frac{30}{7} \approx 4.286$

Exercise 16

Solve for m: $\quad 3.9 = \dfrac{34.2}{m}$.

Solution. Here m sits in a denominator, so first bring it up by multiplying both sides by m:

$$3.9 = \frac{34.2}{m} \implies 3.9\,m = 34.2 \implies m = \frac{34.2}{3.9} = 8.7692\ldots$$

$m = \frac{34.2}{3.9} \approx 8.769$

Exercise 17

Solve for k: $\quad 9.7 = \dfrac{157.4}{\sqrt{k}}$.

Solution. Two steps, in this order: clear the denominator, then undo the square root by squaring.

$$9.7 = \frac{157.4}{\sqrt{k}} \implies 9.7\sqrt{k} = 157.4 \implies \sqrt{k} = \frac{157.4}{9.7} = 16.2268\ldots$$

Now square both sides, using $\left(\sqrt{k}\right)^2 = k$:

$$k = \left(\frac{157.4}{9.7}\right)^2 = (16.2268\ldots)^2 = 263.309\ldots$$

$k = \left(\frac{157.4}{9.7}\right)^2 \approx 263.31$

1.6 Lines

SLOPE-INTERCEPT FORM

$$y = mx + b,$$

where m is the **slope** (the number multiplying x) and b is the ***y*-intercept** (the number standing alone). This is the simplest model in mathematics, and in this course it is the model we will fit to data.

Exercise 18

For the following equations of lines, find the slope and the y-intercept.

(a) $y = 3x + 8$

(b) $y = 16x - 5.67$

(c) $y = -2 + 5x$

(d) $y = 0.00789 - 1.2345x$

Solution. Match each equation against $y = mx + b$. Watch the *sign* attached to each number, and do not be fooled by the order in which the terms are written.

	equation	slope m	intercept b
(a)	$y = 3x + 8$	3	8
(b)	$y = 16x - 5.67$	16	-5.67
(c)	$y = -2 + 5x$	5	-2
(d)	$y = 0.00789 - 1.2345x$	-1.2345	0.00789

In (c) and (d) the two terms are written in the opposite order. The slope is still whatever multiplies x, and the intercept is still the lone number — position on the page decides nothing.

1.7 Evaluating formulas

In this course the work is more about reasoning than about computation. Even so, you must be able to evaluate a new formula correctly on a basic scientific calculator. A good habit: first run the formula with easy numbers you can check in your head, so that you learn the right order of keystrokes. Only then put in the messy numbers.

Exercise 19

Let $l = x + f \cdot d$.

i. Find l when $f = 3,\ d = 2,\ x = 11$.

ii. Find l when $d = -8,\ f = 3,\ x = 11$.

iii. Find l when $x = 8.79,\ d = 1.7,\ f = 3.23$.

Solution. Multiplication happens before addition, so compute $f \cdot d$ first and only then add x.

i. $l = 11 + 3 \cdot 2 = 11 + 6 = 17.$

ii. $l = 11 + 3 \cdot (-8) = 11 - 24 = -13.$

iii. $l = 8.79 + 3.23 \cdot 1.7 = 8.79 + 5.491 = 14.281.$

In (ii) the product is negative, so adding it moves the answer down. If your calculator returned -104 you typed $(11 + 3) \cdot (-8)$: the parentheses were not there.

i. 17 ii. -13 iii. 14.281

Exercise 20

Let $w = \dfrac{x - m}{s/\sqrt{n}}$.

i. Find w when $x = 9,\ m = 6,\ s = 8,\ n = 16$.

ii. Find w when $x = 0.49,\ m = 0.5,\ s = 0.063,\ n = 17$.

Solution. Build the numerator and the denominator separately, then divide once at the end.

i. Numerator $9 - 6 = 3$. For the denominator, $\sqrt{16} = 4$, so $s/\sqrt{n} = 8/4 = 2$. Hence

$$w = \frac{3}{2} = 1.5.$$

ii. Numerator $0.49 - 0.5 = -0.01$, which is negative — that is fine and it means x sits below m. For the denominator, $\sqrt{17} = 4.12310\ldots$, so

$$\frac{s}{\sqrt{n}} = \frac{0.063}{4.12310\ldots} = 0.0152798\ldots$$

and therefore

$$w = \frac{-0.01}{0.0152798\ldots} = -0.65446\ldots \approx -0.654.$$

i. $w = 1.5$ ii. $w \approx -0.654$

CAREFUL

This w is not a random formula: it is the z-statistic we will use all term. Get the keystrokes right now. On most calculators you must write the denominator inside parentheses, `(0.063/\17)`, or the machine will divide only by 0.063 and then multiply by $\sqrt{17}$.

Exercise 21

Let $R = \sqrt{\dfrac{m_1^2}{n_1} + \dfrac{m_2^2}{n_2}}$.

i. Find R when $m_1 = 6,\ n_1 = 4,\ m_2 = 8,\ n_2 = 4$.

ii. Find R when $m_1 = 6.21,\ n_1 = 32,\ m_2 = 8.73,\ n_2 = 49$.

Solution. Everything under the root must be finished before the root is taken.

i.

$$\frac{6^2}{4} + \frac{8^2}{4} = \frac{36}{4} + \frac{64}{4} = 9 + 16 = 25, \qquad R = \sqrt{25} = 5.$$

ii.

$$\frac{6.21^2}{32} = \frac{38.5641}{32} = 1.205128\ldots, \qquad \frac{8.73^2}{49} = \frac{76.2129}{49} = 1.555365\ldots$$

Adding, $1.205128 + 1.555365 = 2.760493\ldots$, and then

$$R = \sqrt{2.760493\ldots} = 1.66147\ldots \approx 1.661.$$

i. $R = 5$ ii. $R \approx 1.661$

CAREFUL

A square root does *not* distribute over a sum:

$$\sqrt{x+y} \neq \sqrt{x} + \sqrt{y}.$$

In part (i), $\sqrt{9+16} = \sqrt{25} = 5$, whereas $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$. Different answers. Add first, then take the root.

Exercise 22

Let $r = \dfrac{m-t}{s}$.

i. Find m when $r = 2,\ s = 3,\ t = 4$.

ii. Find m when $r = 1.84,\ s = 3.12,\ t = 3.1$.

Solution. This time the unknown is not the subject of the formula, so rearrange first. Multiply both sides by s and then move t across:

$$r = \frac{m-t}{s} \implies rs = m - t \implies m = rs + t.$$

Only now substitute.

i. $m = 2 \cdot 3 + 4 = 6 + 4 = 10.$

ii. $m = 1.84 \cdot 3.12 + 3.1 = 5.7408 + 3.1 = 8.8408 \approx 8.841.$

i. $m = 10$ ii. $m \approx 8.841$

1.8 Summation notation

The notation $\sum_{i=1}^{n} x_i$ stands for the sum of n numbers $x_1, x_2, \ldots, x_n$, and is read *the sum of all* x_i *with* i *running from* 1 *to* n:

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \cdots + x_n.$$

The term following the sign $\sum$ (the Greek letter sigma) says *what* is being added; the labels below and above say *how many* terms there are.

Exercise 23

Let x represent the number of M&M's contained in individual trail mix snacks. The data for five trail mix snacks is recorded below.

Trail mix (i)	1	2	3	4	5
M&M's count (x_i)	6	5	6	7	8

(a) The formula that means to sum up the M&M's counts of all five trail mixes is $\sum_{i=1}^{5} x_i$. Evaluate it.

(b) Show the formula that means to sum up the M&M's counts of the first three trail mixes, and evaluate it.

(c) The formula that means to sum up the squares of all M&M's counts is $\sum_{i=1}^{5} x_i^2$. Evaluate it.

(d) The formula for the square of the sum of all M&M's counts is $\left(\sum_{i=1}^{5} x_i\right)^2$. Evaluate it.

(e) Fill in the blank with the proper symbol, $=$ or $\neq$: $\displaystyle\sum_{i=1}^{5} x_i^2 __ \left(\sum_{i=1}^{5} x_i\right)^2$.

Solution. **(a)** Add the five counts:

$$\sum_{i=1}^{5} x_i = x_1 + x_2 + x_3 + x_4 + x_5 = 6 + 5 + 6 + 7 + 8 = 32.$$

(b) "The first three trail mixes" means stopping at $i = 3$. What is being added is still x_i, the counts — not the labels i:

$$\sum_{i=1}^{3} x_i = x_1 + x_2 + x_3 = 6 + 5 + 6 = 17.$$

(c) Now each count is squared *before* adding:

$$\sum_{i=1}^{5} x_i^2 = 6^2 + 5^2 + 6^2 + 7^2 + 8^2 = 36 + 25 + 36 + 49 + 64 = 210.$$

(d) Here the adding happens first and the squaring last. Using the total from (a),

$$\left(\sum_{i=1}^{5} x_i\right)^2 = 32^2 = 1024.$$

(e) Comparing (c) and (d), 210 against 1024:

$$\sum_{i=1}^{5} x_i^2 \neq \left(\sum_{i=1}^{5} x_i\right)^2.$$

(a) 32 (b) 17 (c) 210 (d) 1024 (e) $\neq$

CAREFUL

Parts (c) and (d) look almost the same written down, and they are not the same number at all: the sum of the squares is 210, the square of the sum is 1024. Square-then-add and add-then-square are different instructions. Both appear in the formula for the standard deviation, so the distinction will cost you marks if you let it slide now.